

MAL Data Flow & Pipeline Dependencies

Mock / Development Data Pipeline

Mock Raw Inputs

DATA_6/KETI/*/*.csv
(Distributed Sensors)

DATA_7/HomeCoach_5min_*.csv
(HomeCoach CSVs)

DATA_1/room_conditions.csv
(Room Occupancy)

DATA_8/.../1_room_measurements.csv
(Smart Campus)

Resampling & Preparation

«Script»
resample_keti_data.py

«Script»
prepare_homecoach.py

«Notebook»
1_data_preparation.ipynb
2_source_datasets_analysis.ipynb

Interim Mock Data

keti_1min_resampled.csv

HomeCoach_combined.csv

Merging & Unifying

«Script»
build_unified_environment_dataset.py

Unified Dataset

unified_environment_focus_dataset.csv

Unified Dataset Cleaning & Imputation

«Notebook»
3_unified_dataset_analysis.ipynb

«Script»
clean_and_impute_dataset.py

Cleaned Unified Dataset

final_mock_dataset.csv

Session Linearization

«Notebook»
4_session_linearization.ipynb

«Script»
linearize_session_windows.py

Linearized Intermediate

linearized_sessions_with_missing_targets.csv

Target Imputation

«Notebook»
5_fill_missing_targets.ipynb

«Script»
fill_missing_targets.py

Model-Ready Mock Dataset

linearized_session_windows.csv

Training & Artifacts

«Script»
train_model.py

rf_model.pkl

Real Production Pipeline

Production Data Source

PostgreSQL DB
(Real Production Data)

Data Extraction

«Script»
extract_data.py

Database Exports

real/sensor_history.csv
real/sessions.csv

Real Linearization & Preprocessing

«Notebook»
6_real_data_analysis_and_preprocessing.ipynb

«Script»
linearize_and_glue.py

Real Model-Ready Dataset

real/linearized_sessions_with_target.csv

Color / Line style	Description
	Python Scripts (Production / Automated pipeline)
	Jupyter Notebooks (Exploration / Redundant logic)
	CSV Files
#blue;line.dotted	Triggers: Notebook triggering production script (Zero conflict risk)